Department of Nano Science Technology

Faculty of Technology

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Centralized Workshops and Laboratory Allocation Platform

User Manual

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Module: NANO2142

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1. Install and launch

- Extract the complete SYNCLAB package into a writable folder.

- Open B6 / Windows_x64 and double-click Run_SYNCLAB_Group_01.cmd.
- The launcher starts the bundled 64-bit application using data/student_data.db. A separate Qt installation is not required for this launch method.
- For a source build, open B5 / SYNCLAB / StudentWorkshopApp.pro using a Qt 6 Desktop kit.

2. Welcome screen

The Welcome screen provides three options: New Allocation, Edit Allocation and Settings. Choose the option that matches the task.

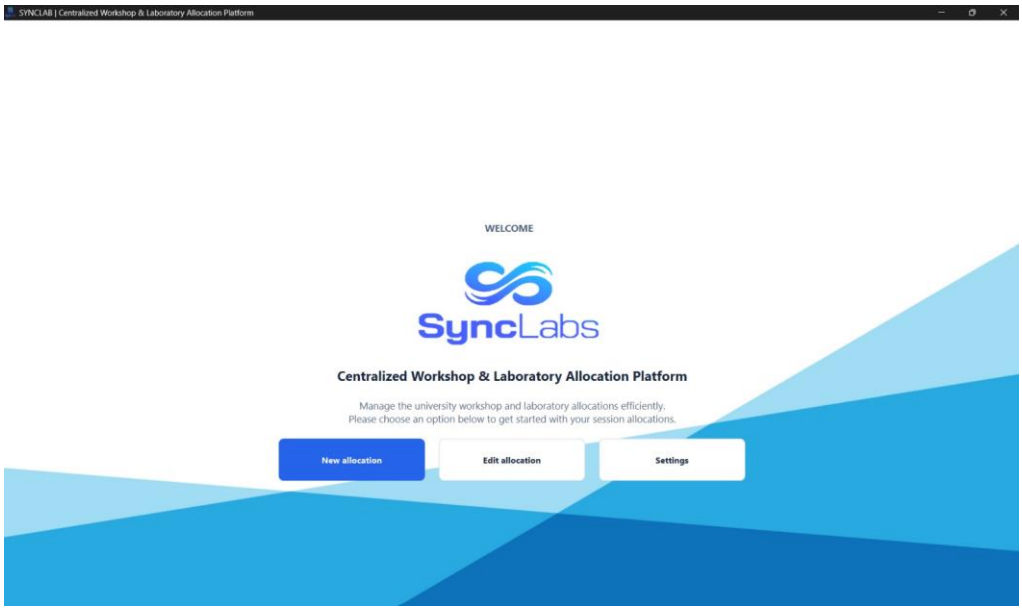


Figure 1. SYNCLAB Welcome screen

3. Create a new allocation

1. Select New Allocation and choose the required laboratory or workshop.
2. On the calendar, choose today or a future date. Select one or more available time slots. Disabled slots are reserved or belong to a past date.
3. Enter your name, index number, department, email address, supervisor name and project name. Description is optional.
4. Review the details and confirm the allocation.
5. Keep the displayed four-digit Allocation Key. The request is saved with Pending status until it is reviewed.

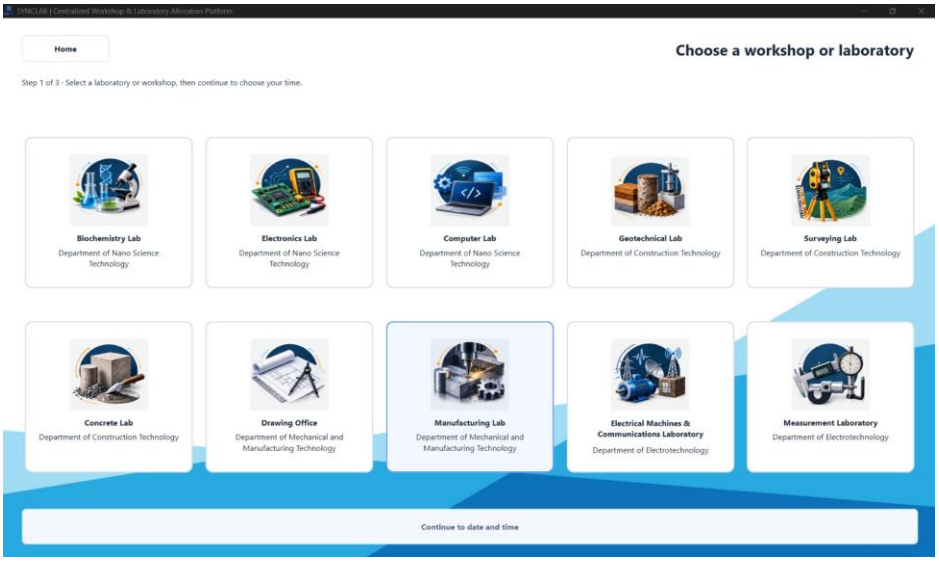


Figure 1 Workshop and Laboratory screen

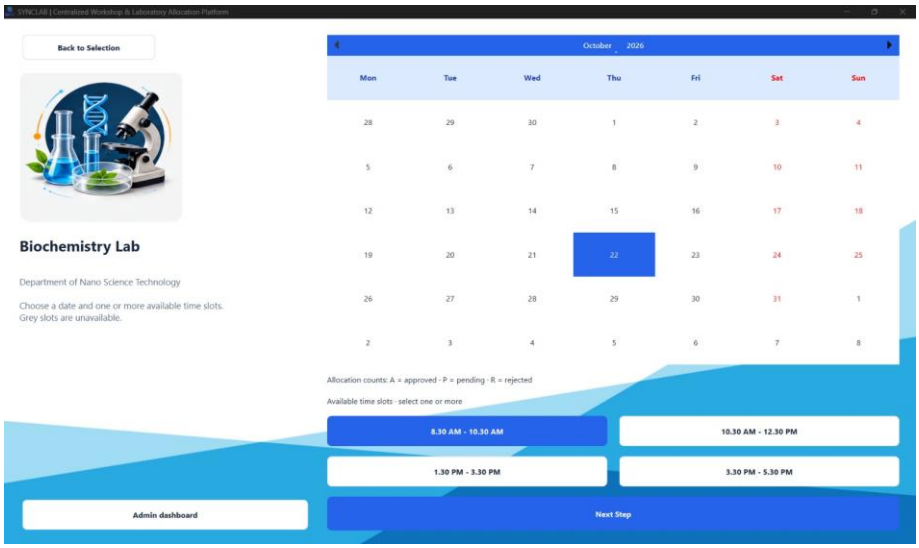


Figure 2. Calendar and time-slot selection screen

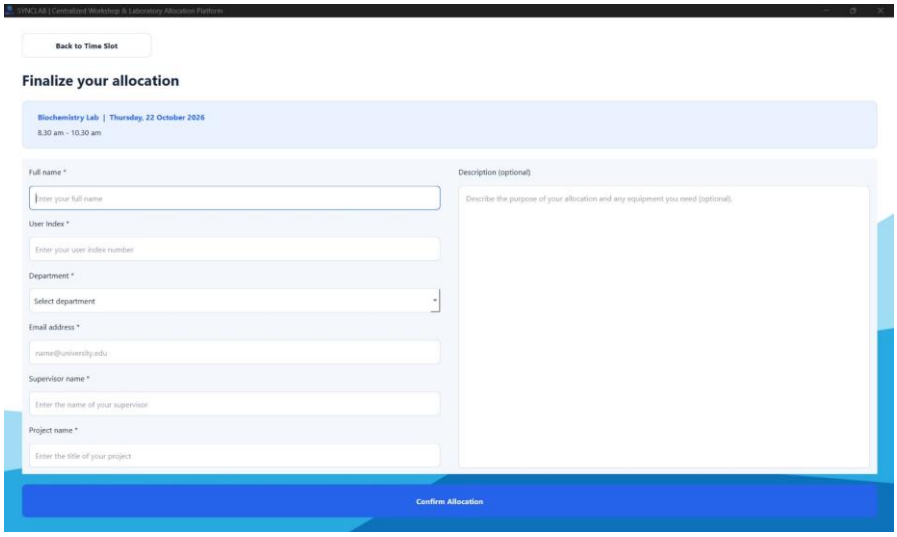


Figure 2 Personal details and project details

4. Edit or cancel an allocation

6. From Welcome, select Edit Allocation and enter your saved Allocation Key.
7. If the system requests an index number for a legacy duplicate, enter your student index too.
8. Review the saved details. You can change the laboratory, future date, available slots or form information.
9. Select Update Allocation to save changes. The key remains the same and the request returns to Pending.
10. Select Delete Allocation to cancel the request; confirm when prompted. Past bookings can be viewed or deleted, but updates require a current or future schedule.

5. Administrator access and review

11. Start the new-allocation path, select the relevant laboratory and open Admin Dashboard from its calendar.
12. Enter the laboratory's administrator passcode. The initial passcode is 0000 only if no replacement has been saved in local settings.
13. Choose a date to filter the records, or select Show All Dates to remove the date filter.
14. Select a row before using Edit, Delete or an email action. Scroll the table horizontally to view additional columns, including remarks and status.
15. Change the status using the status control: Pending, Approved or Rejected. Remarks are saved when the editor loses focus.

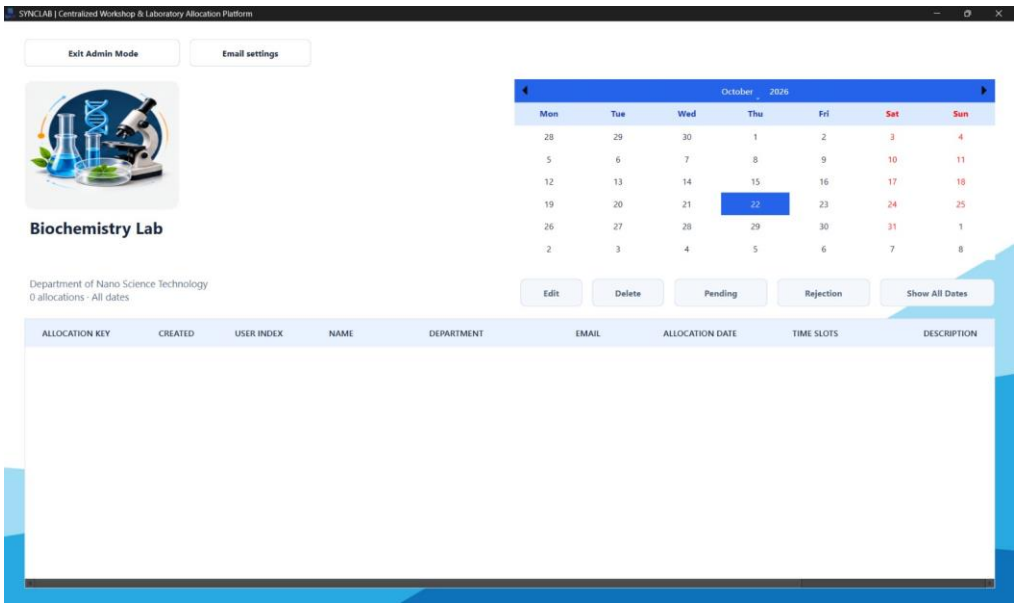


Figure 3. Administrator dashboard and allocation table

Status behaviour: Pending and Approved requests reserve their slots. Rejected requests release their slots. Re-approving a request checks that the slot is still available. An administrator edit keeps the previous status.

6. Approval and rejection email drafts

- Configure the in-charge and department/head email addresses in administrator email settings.
- Choose the relevant allocation and use Pending or Rejection to prepare the email draft.
- The draft opens in the computer's default email application. Review and send it manually. SYNCLAB does not send emails or confirm delivery.
- If no email handler is available, a warning may appear; the saved allocation status remains.

7. System settings

- Select Settings from the Welcome screen and enter the master passcode. The initial default is 0000 unless it has been changed.
- Available settings include workshop/master passcodes, light or dark mode, and a wallpaper image.
- Keep the wallpaper file available because the application saves its file path.
- Settings and notification email addresses are stored in QSettings, separately from the allocation database.

8. Back up and restore data

Back up

16. Close SYNCLAB completely before copying the database.
17. For the supplied launcher, copy Windows_x64/data/student_data.db to a dated backup folder.
18. For source builds, confirm which data folder is being used before copying the database. Keep configuration details and any external wallpaper separately.

Restore

19. Close the application and keep a separate copy of the current database.
20. Replace the selected student_data.db with a compatible backup copy.
21. Reopen SYNCLAB and verify record counts, dates, Allocation Keys and statuses.

Backup and restore are manual file operations. During a detected legacy migration, the application also creates a .pre-4nf.bak copy.

9. Important notes and limitations

- Keep the runnable folder intact and launch the application using its supplied launcher.
- The workshop list and four daily time-slot periods are fixed. The system does not model facility capacity or closures.
- Allocation Keys and passcodes are stored in plain local data/settings and provide limited access protection. Change the initial administrator defaults.
- There is no active PDF upload, built-in report export, in-app recovery or remote synchronization.
- Changing a profile that uses the same student index affects its joined details across allocations.
- Keep real records separate from the supplied fictional demonstration data.